

Nonbank Market Power in Leveraged Lending

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Abstract. Banks finance their lending to risky firms by selling these loans to nonbank financial institutions. Among these nonbanks, collateralized loan obligations (CLOs) provide the bulk of funds. I show that CLO managers have significant market power, which enables them to extract lender-friendly loan terms. Their market power results from switching costs faced by the bank. One source of switching costs is information asymmetries across CLO managers that arise during underwriting. To identify my results, I construct a new instrument using novel data on mergers in the CLO industry. I provide the first analysis of these mergers and their determinants.

Keywords: Nonbank Lending, Leveraged Loans, Loan Underwriting, CLOs, Market Power

JEL Classification: G14, G21, G23, G24, G34

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I Introduction

Today many corporate loans seem to resemble bonds in all but name. Banks underwrite loans with bond-like features to sell to nonbank financial institutions, many of which also invest in the bond market. This development has not gone unnoticed. In a recent legal case, which the loan market’s advocacy group warned will “jeopardize a trillion-dollar market that is vital to the economy,” the U.S. Court of Appeals had to answer whether these loans should in fact be treated as public securities. Despite the apparent convergence of loan and bond markets, loans continue to carry perplexingly large spreads compared with their bond counterparts ([Schwert, 2020](#)). The credit spread on an otherwise identical loan is nearly twice that of a bond.

My paper establishes that imperfect competition in the loan market is an important contributor to high loan spreads. However in doing so, I depart from a long literature that studies inter-bank competition. Instead, I turn to competition in the nonbank sector and provide evidence on the market power of nonbanks, and collateralized loan obligations (CLOs) in particular.

In addition to showing that CLO managers have significant market power, I also investigate the source of this market power. I argue that information asymmetries arise during loan origination, because loans are not public securities. These information asymmetries create an “information oligopoly” for CLO managers, which resembles the information monopoly enjoyed by incumbent banks in traditional lending ([Rajan, 1992](#); [Sharpe, 1990](#)). Because of the hold-up problem associated with this information oligopoly, it is costly for the bank to replace the funds of one CLO manager with the funds of another CLO manager.¹

As a third contribution, I provide the first analysis of mergers in the CLO industry. I show that these mergers are liability-driven, not asset-driven. Rather than pursuing targets for their loan portfolios, acquirers choose targets to improve their CLO funding conditions. This form of target selection, combined with targets’ high levels of diversification, make the acquired loans random additions to the acquirer’s portfolio. This justifies my use of these additions as an instrument in my main analysis.

¹Some work refers to the rent extraction due to an incumbent bank’s information monopoly as “market power,” while other work refers to it as “bargaining power.” I use the former terminology.

I establish these results in the context of the leveraged loan market. The term “leveraged loan” refers to a loan to a sub-investment-grade borrower, which is typically syndicated. While banks continue to provide capital for credit lines, nonbank lenders have largely replaced banks in the provision of capital for term loans. This market for institutional term loans has grown rapidly since the Global Financial Crisis. Today nonbank loans constitute the primary source for high yield corporate debt in the United States. To illustrate, in 2024 risky firms raised \$1.22 trillion in institutional term loans compared to \$233 billion in bonds (White & Case, 2025).

Regulatory capital requirements make it costly for banks to hold risky loans (Irani, Iyer, Meisenzahl, and Peydró, 2021). Consequently, the transfer of these loans from a bank’s balance sheet to the balance sheets of less regulated institutions can mutually benefit both parties. Typically, the underwriting agreement prevents the arranging bank from freely adjusting a loan’s terms to guarantee the sale of the entire issue. Thus, an arranger who is unable to sell a loan in full may be required to retain a loan’s unsold portion. This risk to banks from loan underwriting has been coined “pipeline risk” (Bord and Santos, 2015; Bruche, Malherbe, and Meisenzahl, 2020).

Nonbanks that account for significant demand for a given loan may internalize the fact that their lending decision represents a risk to the arranging bank. This may grant market power to the nonbank to command lender-friendly loan terms. However, the underwriting agreement aligns the arranging bank’s incentives with those of the borrowing firm by remunerating the arranger for borrower-friendly terms through increased underwriting fees. Thus, the arranger may have to choose between the cost of retaining a larger loan share and the reduction in underwriting fees associated with giving in to the nonbank.

For a given new loan issue, I measure a nonbank’s market power over the arranging bank as the dollar volume of the nonbank’s holdings in the borrowing firm’s pre-existing loans. These holdings reflect the nonbank’s demand for the firm’s outstanding loans, which is informative about her demand for the new loan issue. Thus, my measure captures the extent to which the nonbank exposes the arranging bank to pipeline risk.

The complication with this measure is that for any investor a large holding may simultaneously reflect positive private beliefs about the borrower’s quality. These beliefs can for

example arise as a result of private information. The anticipated effect of private beliefs is in the opposite direction of the effect of market power. While the market power from large holdings would lead to a higher spread, positive private beliefs may instead induce a nonbank to accept a lower spread. Thus, failure to account for private beliefs may negatively bias my estimates of the effect of market power.

To empirically disentangle these two forces, I employ exogenous portfolio variation arising from mergers in the largest class of nonbank investors, namely CLOs. A CLO is an actively managed loan fund financed with a tranching debt structure. CLO managers, who are often affiliated with large private equity firms, such as KKR, or insurance companies, such as Prudential Financial, typically manage many CLOs. A CLO’s management contract aligns the CLO manager with the CLO’s equity class. In total, this sector provides 63% of primary market funds for leveraged term loans. For comparison, the next largest primary market investor category are mutual funds with 13%, followed by banks with 12% ([Hinckley et al., 2022](#)). While the previous discussion applies to nonbank investors broadly, I focus on CLOs because they provide frequent and detailed portfolio disclosures necessary for studying primary market transactions.²

My main specification compares the terms at which a given CLO manager lends to different borrowers in the same industry, controlling for time-varying credit risk. Recall that the influence of a CLO manager’s private beliefs about the borrower counteracts the market power effect. Thus, my identification strategy instruments a CLO manager’s holdings of a borrower with the amount of that borrower’s loans obtained through her M&A activity. The main identifying assumption is that merger target portfolios are unrelated to the acquirer’s beliefs about borrowers within an industry. I justify this assumption below.

Using this instrument, I show that CLO managers who expose the bank to pipeline risk indeed have market power to affect loan prices. A one-standard-deviation increase in a CLO manager’s holdings enables her to command an 13.3 basis point higher credit spread and a 6.5 basis point higher original issue discount. Taken together the borrower experiences an increase in cost of debt of 15.5 basis points. These numbers constitute a significant increase in

²Legally, primary market sales to CLOs are structured as secondary market transactions. In practice, these transaction are contracted at primary market terms prior to a loan’s closing. Therefore, economically, they represent primary market acquisitions.

a borrower's funding costs comparable to market power effects measured in the relationship banking literature.

Nonbank market power is also evident in secondary market prices. On average a loan's initial secondary market price lies above its primary market price. In other words, loans are underpriced at issuance, like initial public offerings of equities. I show that the extent of loan underpricing increases with my measure of nonbank market power. This pattern in post-issuance prices represents additional evidence that my estimates reflect market power rather than compensation for risk. If higher spreads were driven by omitted risk factors, the secondary market would price those risks.

Next, I turn to the mechanism that yields market power to CLO managers. A necessary condition for market power is that the bank cannot costlessly replace a nonbank's funds with other capital. Therefore, I ask, what impedes banks from substituting for a nonbank's funds?

Motivated by the importance of private information in traditional bank lending ([Rajan, 1992](#); [Sharpe, 1990](#)), I turn to the role of information in institutional loan underwriting. I argue that information asymmetries *across* nonbanks create a hold-up problem for the bank, which makes it costly for the arranging bank to switch between them. In the presence of information asymmetries, nonbank investment decisions serve as informative signals about borrower quality. Thus, the loss of a previously highly invested CLO manager may depress demand for a new issue through revised nonbank beliefs.

Information asymmetries across nonbank investors arise during syndication. The arranging bank collects private information about the borrowing firm. During book building the bank distributes information to potential investors in the form of memoranda, also called bank books. Different from securities underwriting, the bank assembles both a private bank book containing material private information about the borrower and a public bank book free of such information. Many nonbank investors opt for the public bank book and restrict their information access to avoid jeopardizing their public securities trading with the risk of insider trading allegations.

Importantly, I observe information asymmetry effects in loan pricing. Consistent with the arranging bank experiencing a hold-up problem, market power effects are more pronounced for

high information asymmetry borrowers. For example, loans of borrowers for which secondary markets are less informative because their outstanding loans did not trade recently are more than twice as sensitive to market power. For these borrowers a one-standard-deviation move in market power by my measure increases the cost of borrowing by 41.4 basis points. For comparison, a move in credit rating from B+ to B is accompanied with an average borrowing cost increase of 44.4 basis points in my sample.

Lastly, I return to the question of my instrument's validity. The principal threat to my identification is that merger-obtained holdings correlate with acquirers' private beliefs about borrower credit quality. In particular, acquirers may select targets with loan portfolios that reinforce their tilt towards their preferred assets. Note that portfolio-based target selection that is independent of beliefs does not undermine my identification. One such motive is diversification.

To address concerns about belief-based target selection, I argue that during my sample period acquirers select targets based on their CLO liabilities, not their assets. Specifically, acquirers target CLO managers who are constrained in their ability to issue new CLOs due to risk retention requirements or whose CLOs are financed with low leverage or expensive debt tranches. The value proposition for merging stems from both the ability to back the acquired assets with CLO debt refinanced at improved terms and from acquirers' capacity to provide equity for new CLO issues.

I demonstrate that, in terms of their assets, targets are indistinguishable from alternative, but non-selected, CLO managers. Instead it is the structure of their liabilities that distinguishes targets. More precisely, targets are more likely than non-targets to manage CLOs with low leverage or high cost of debt. Public statements on merger rationales corroborate these results. In nearly nine of ten mergers, participants point to improved capital access either through available equity or outside investors. Less than 10% of merger statements contain any reference to the loan portfolio, with all such statements being general, not borrower-specific. Regardless, I take a conservative approach and exclude these mergers from my sample.

This article contributes to several literatures. Many ideas are drawn from the study of the competitive effects of information in the classic bank-firm model tracing back to the seminal

theoretical insights of [Sharpe \(1990\)](#) and [Rajan \(1992\)](#). [Schenone \(2010\)](#) provides evidence of bank’s information monopolies, [Paravisini, Rappoport, and Schnabl \(2023\)](#) find that lender specialization amplifies informational advantages, and [Darmouni \(2020\)](#) finds that adverse selection increases the cost of lending for non-relationship banks.³ I find that banks face an information problem with respect to nonbanks. In some ways banks and nonbanks in my setup resemble borrowers and banks in the classic model.

Methodologically, my use of merger-induced variation in lender portfolios draws on a line of work in banking that employs mergers as an instrument to separate market power from private lender information (e.g., [Erel, 2011](#); [Favara and Giannetti, 2017](#); [Garmaise and Moskowitz, 2006](#); [Giannetti and Saidi, 2019](#); [Saidi and Streitz, 2021](#); [Sapienza, 2002](#)). I depart from this literature in two important ways. First, I employ mergers between nonbanks rather than between banks. Second, I focus on competition from nonbank lenders, not between banks.

Lastly, my work contributes to a burgeoning literature on the role of nonbanks in corporate lending. [Chernenko, Erel, and Prilmeier \(2022\)](#) show that firms with intangible or risky assets borrow directly from nonbanks precisely when traditional banks are unwilling to lend.⁴ I complement these studies by showing that strategic interactions of nonbanks can influence the loan terms even when arrangers act as intermediaries between nonbanks and borrowers.

Within work on the leveraged loan market, [Irani et al. \(2021\)](#) and [Irani and Meisenzahl \(2017\)](#) link nonbank growth to bank capital regulation and liquidity, while [Ivashina and Sun \(2011\)](#) and [Fleckenstein, Gopal, Gutiérrez, and Hillenbrand \(2025\)](#) show the influence of nonbank loan demand on loan terms. Specifically for CLOs, [Bhardwaj, Ge, and Mukherjee \(2022\)](#) turn to the insurance sector to show that demand for CLO debt tranches affects CLO formation and in turn loan market conditions. Their finding that CLO issuance has important consequences for CLO managers and borrowers is reflected in my results that CLO liabilities are the dominant determinant of mergers in this industry.

The paper most closely related to my own is [Bruche et al. \(2020\)](#). These authors study the

³A separate literature studies the effect of competition among uninformed banks. See, e.g., [Petersen and Rajan \(1995\)](#), [Boot and Thakor \(2000\)](#), and [Cetorelli and Strahan \(2006\)](#).

⁴[Davydiuk, Marchuk, and Rosen \(2024\)](#) and [Erel, Flanagan, and Weisbach \(2024\)](#) provide parallel evidence on business development companies and private debt funds.

demand discovery problem faced by arranging banks in the underwriting markets. In their model an arranging bank sells a loan to a representative nonbank who is better informed than the bank about the value of the loan. To deter the nonbank from indicating low demand for high quality loans, the truth-revealing mechanism requires the bank to retain a larger share of the loan whenever the nonbank indicates low demand. My analysis complements the important insights from their work in a number of ways. First, I deviate from the representative investor assumption and instead explore heterogeneity among nonbanks. Second, this allows me to focus on another dimension of information asymmetries. By focusing on the information asymmetries across nonbanks, I sidestep the issue of whether nonbanks are indeed better informed than the bank and show that their notion of pipeline risk is relevant without this assumption. Third, their work tends to focus on bank outcomes and employs loan price adjustments to explain bank retention. In contrast, my work predominantly studies the impact of market power on loan prices and thus concerns borrower outcomes.

The remainder of this article is structured as follows. Section III presents the data and Section II gives some institutional background. Section IV investigates the extent to which CLO managers have market power. Section V studies the source of CLO managers' market power. Section VI discusses the determinants of mergers in the CLO industry and provides evidence on the instrument's validity. Section VII concludes.

II Institutional Background

The leveraged loan market first received the attention of regulators at the start of the 1990s due to concerns over commercial banks' involvement in "highly leveraged transactions." Since then, supervisory scrutiny and risk-based capital requirements have imposed significant regulatory costs on banks for high-yield corporate debt held on their balance sheets. In response, banks largely retreated from funding term loans of large, risky firms. Nonbank financial institutions filled this gap and have since become the dominant source of capital for term loans in the leveraged loan market.

Today, institutional loan issuance outstrips that of the bank-funded market segment. In total, institutional loan volumes have exceeded those of bank term loans sevenfold since the Global Financial Crisis. During that period, the institutional loan market has even surpassed

its public-market counterpart, with close to double the issuance volumes of high-yield bonds (Figure 1). As annual issuance approaches one trillion dollars, the institutional loan market has become the principal source of debt capital for sub-investment grade corporate borrowers in the United States.⁵

Despite these structural changes, banks remain central to the modern leveraged loan market. As reliance on traditional commercial bank lenders declined, the market transitioned to an underwrite-to-distribute model. Rather than fund loans directly, banks now primarily fulfill an investment bank function. In their capacity as loan arrangers, banks market loans on behalf of corporate issuers to nonbank lenders.

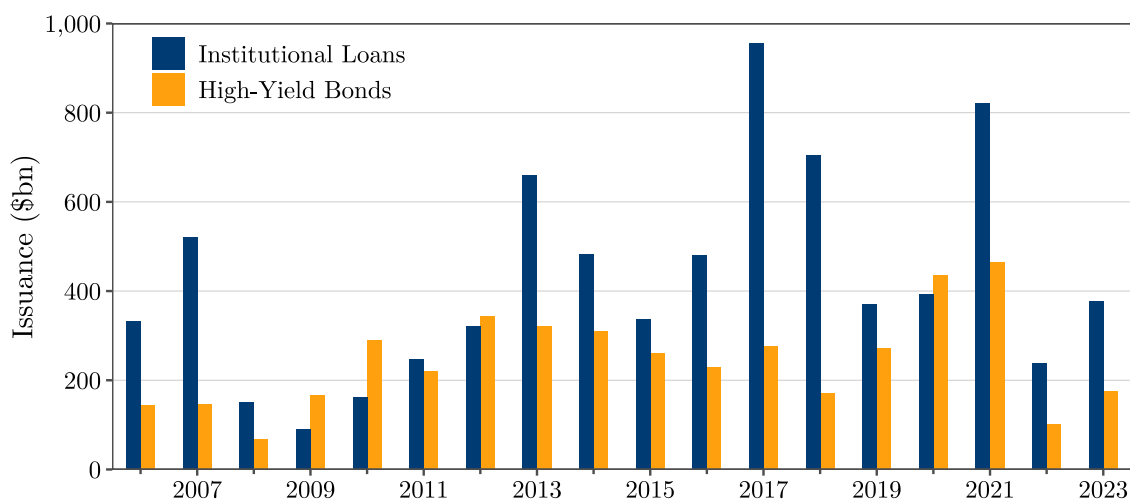


Figure 1. U.S. Sub-Investment-Grade Corporate Debt Issuance

This figure shows annual U.S. institutional loan and bond issuance volumes by sub-investment grade corporate borrowers from 2006 to 2023. Issuance data comes from LCD.

To compete for a borrower’s loan mandate, banks propose preliminary terms and conditions which form the foundation of the subsequent negotiation between the borrower and the selected arranger. At the end of their negotiation, a term sheet contractually formalizes loan pricing and other key contract terms. Importantly, at this time, the arranger of an underwritten deal guarantees that the borrower will receive the full loan amount after syndication. Only then does the arranger begin to market the loan.

⁵Figure IA.A1 contrasts annual issuance volumes of institutional loans with bank term loans in the leveraged loan market. Figure IA.A2 shows market shares of sub-investment grade term debt markets over time.

This sequence of events exposes the arranger to substantial underwriting risk. Because contract terms are agreed upon before demand discovery, loan terms are negotiated based on anticipated rather than realized demand for a loan. Consequently, market clearing terms may deviate from those initially negotiated due to misjudged demand expectations during negotiation or due to changes in market conditions during syndication. If such ex-post loan mispricing precludes syndication of the entire loan, the loan amount guarantee made to the borrower requires the arranger to fund any unsold remainder. Providing these funds directly is costly for the arranger because of the high regulatory capital charges associated with these risky loans.

To reduce this pipeline risk, the arranger is permitted to adjust loan pricing, and occasionally other terms, during syndication. However, the arranger must forfeit a portion of their underwriting fees in return when they make lender-friendly loan term adjustments. By ensuring that the arranger participates in the increased cost of borrowing, this risk-sharing agreement prevents the bank from simply changing a loan’s terms against the borrower and helps to align the arranger’s incentives with those of the borrower.

III Data Sources

My analysis primarily draws on data from four sources: CLO security-level portfolio holdings data from LSEG LPC Collateral, CLO issuance and liabilities data from PitchBook | LCD, hand-collected information on mergers and acquisitions (M&A) in the CLO industry, and loan origination data from LSEG Loan Connector. I reference data from other sources in the analysis.

First, I obtain CLO security-level portfolio holdings data from LPC Collateral which LSEG collects from CLO trustee reports. For quarters in which a CLO has multiple reports available, I include only data from the last report. Besides holding amounts, I retrieve current credit ratings issued by Moody’s, S&P, and Fitch. For comparability, rating references are made at the S&P scale.

Second, I employ CLO issuance data from LCD. I link CLOs in LPC Collateral to LCD using a fuzzy string match and individually verify the match validity. For the unmatched remainder, I manually supplement matches.

Third, I hand-collect data on events that result in the transfer of CLO management responsibilities. Formally, a CLO manager is the legal entity to which a CLO assigns portfolio management rights per its collateral management agreement. Transactions that transfer such rights follow either a management replacement deal or a platform deal structure. In a platform deal, the acquirer obtains control rights over a CLO manager. Thereby, the acquirer implicitly assumes the management rights of its new subsidiary without changes to the acquired manager's collateral management agreements. Alternatively, in a replacement deal, collateral management agreements are amended to assign management rights from the current manager to a new replacement manager. In contrast to platform deals that confer management rights of all CLOs of a manager, replacement deals allow for a subset to be transferred. However, the amendment of each CLO's collateral management agreement requires the separate approval of its noteholders.⁶

To collect a historic record of M&A transactions, I begin with the set of CLO managers at issuance captured by LCD. For each manager, I search their available historic press releases for public statements referring to such transactions. I complement my results with a news search of CLO industry publications, major general financial news outlets, and LexisNexis. Lastly, I identify transactions from SDC's M&A database. This search yields platform deals and large replacement deals. Small replacement deals frequently lack public statements or news coverage. For these deals, I utilize that amendments of collateral management agreements may trigger a credit rating review and collect transactions from associated announcements by Moody's and Fitch.

Fourth, I collect leveraged loan origination data from Loan Connector. LSEG loan identification numbers (LINs) provide a common loan identifier between the origination data and the CLO holdings data. For missing loan characteristics, I supply the corresponding information recorded in LPC Collateral.

I study corporate entities at their highest level of aggregation. This choice accounts for joint

⁶Rarely, replacements occur for reasons other than acquisitions. One reason can be the trigger of a key-man provision. For example, in 2005, four of the six CLOs managed by Kohlberg & Co subsidiary Katonah Capital allowed noteholders to redeem upon a key-man's departure unless Kohlberg proposed a suitable replacement manager. Even though Kohlberg and Allied Capital had previously agreed on the sale of Katonah, the trigger of this provision forced Kohlberg to find new managers. Ultimately, Blackstone and two other managers replaced Katonah. Allied Capital terminated the merger. For details see institutionalinvestor.com/article/b150nn7x1zbz75/a-key-womans-leverage.

loan investment decisions by CLO manager subsidiaries of the same parent. Specifically, some CLO managers established subsidiaries in response to risk retention requirements which the parent controls through shared services and staff agreements. Corporate ownership information stems from S&P NetAdvantage, the CLO manager’s website, and references made in industry publications. I obtain further parent identities as a byproduct of collecting my M&A sample. I aggregate public corporate borrowers with the help of the [Chava and Roberts \(2008\)](#) Compustat link table.

IV The Market Power of CLO Managers

Nonbank financial institutions have largely replaced banks as providers of term loan funding in the leveraged loan market. Nonetheless, banks remain central to this market in their capacity as loan arrangers. An important function of arrangers is to take on a deal’s market risk by guaranteeing the loan before syndication to institutional investors. This guarantee can be costly because arrangers have only limited ability to pass on increased borrowing costs to reflect changed market conditions.

However, unless the arranger can freely compensate for a shortfall of funds from one nonbank with funds from another, even a nonbank’s individual investment decision can represent a risk to the arranger. If a nonbank lender recognizes that their investment decision imposes an externality on the bank, the nonbank may be able to command more favorable loan terms. These represent an economic rent to the nonbank. Thus, I refer to this capacity of the nonbank to attain more lender-friendly loan terms as market power.

The amount of market power that a nonbank can exert depends on the extent to which a loan’s funding is exposed to the nonbank’s lending decision. This funding risk is captured by the size of the shortfall that the nonbank would induce if the nonbank withheld anticipated funds. Therefore, the expected loan demand from a nonbank, as anticipated by the arranger during negotiations with the borrower, measures the risk from that nonbank.

Many factors can influence an investor’s loan demand, which generally makes demand difficult to predict. For instance, major factors include current market conditions, the investor’s overall portfolio size, information disclosed about the firm, and investment restrictions faced by the investor. However, for a borrower with outstanding loans, the arranger has access to

valuable additional information about loan demand: the nonbank’s current holdings of the borrower’s previously issued loans.

The reason is that nonbank lending is highly persistent for a given firm. Figure 2 illustrates this for CLO managers, which are my nonbanks of interest. This scatter plot maps a CLO manager’s holdings in a borrower’s loans before the syndication of a new loan against the manager’s subsequent primary market investment in the new issue, i.e., their primary allocation. The distinct positive relationship between these variables indicates strong explanatory power of prior holdings for the intensive margin of CLO managers’ lending. To incorporate the extensive margin of lending decision, I further include a binned scatter plot. Its first bin contains observations without prior holdings. The other 19 bins split the remaining sample into equal-sized groups. Importantly, each bin contains observations both with and without primary market allocations.⁷ Overall, the inclusion of CLO managers that do not invest attenuates the relationship only modestly, which implies a high probability of continued investment by CLO managers that have prior holdings. In contrast, CLO managers without existing exposure to the borrower provide almost no funds on average.

Because of the presence of an active secondary loan market, a nonbank’s holdings of a loan reflect their current demand for that loan. While frictions, such as trading costs or institutional constraints like CLO covenants, mean that holdings may reflect demand only approximately, a nonbank will trade a loan if their demand absent frictions sufficiently deviates from their holdings. A borrower’s outstanding loans and their new loan are similar debt instruments since they represent promises by the same firm secured by the same assets. Hence, nonbanks’ revealed demand for outstanding loans provides valuable information about nonbank funding for a new issue.

In Internet Appendix D, I formally examine the pattern displayed in Figure 2. I find strong statistical support of lending persistence. In aggregate, CLO managers with existing exposure to the borrower account for more than 83% of CLO funding in my sample. On the extensive margin, the probability that a manager invests in a new loan increases by approximately 50 percentage points if the manager already holds loans of the borrower. Similarly, the elasticity of the investment amount with respect to existing holdings is 0.525. These magnitudes remain nearly unchanged when I compare the behavior of different managers

⁷Individual observations without primary allocation are not discernible as they cluster on the x-axis.

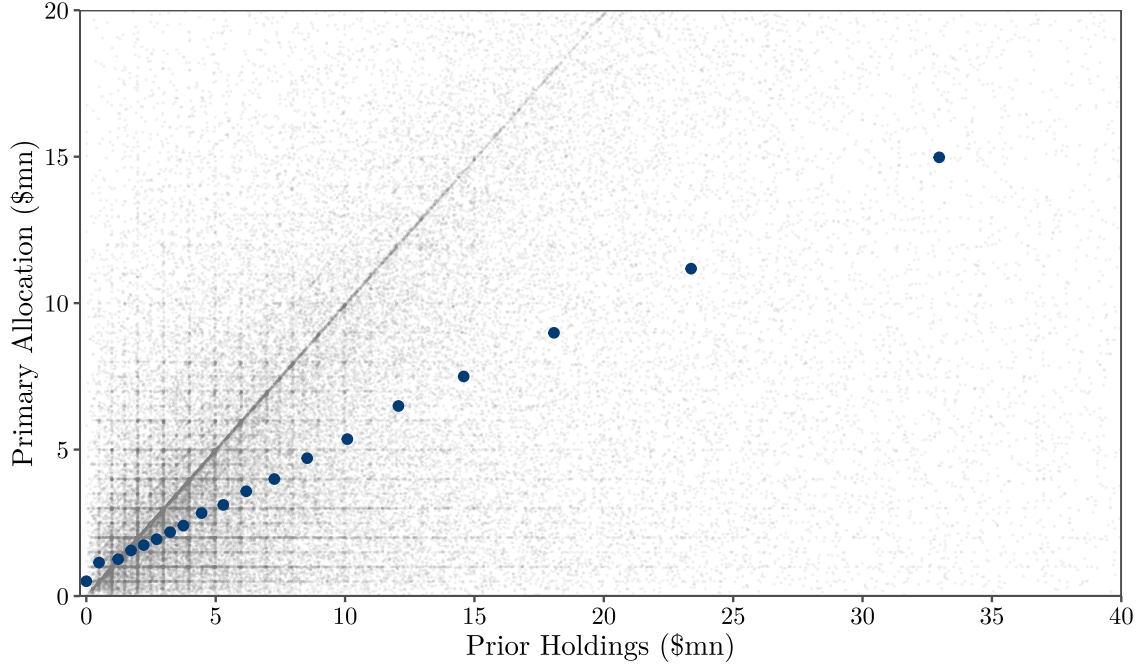


Figure 2. CLO Manager Lending Persistence

This figure shows the relationship between CLO managers' holdings in a borrower's outstanding loans before syndication of a new loan and their primary allocation in the new issue. Each gray dot corresponds to a CLO manager-loan observation in the sample outlined in Table IA.D1. A binned scatter plot is displayed in blue. Its first bin contains observations without prior holdings, i.e., $Prior\ Holding_{f(l)m} = 0$. The remaining 19 bins equally split the positive sample by size.

for the same loan in Table IA.D1. Thus, differences in loan characteristics do not drive this result.

Moreover, CLO managers' lending persistence is not the result of managers merely rolling over their holdings in refinancing deals. While managers are more likely to reinvest refinanced funds, the results in Table IA.D3 establish that holdings strongly predict lending decisions even when a manager's funds are not being refinanced. Lastly, Table IA.D2 documents that lending persistence is also not explained by the presence of common relationship banks that form relationships with both corporate borrowers and nonbanks.

In summary, a nonbank's investment decision imposes a cost if an arranger cannot freely substitute between funds of different nonbank lenders. This potential externality may enable the nonbank to command more lender-friendly loan terms. Although the extent to which a loan relies on funding from any specific nonbank is unobservable, a nonbank's prior holdings

of the borrower’s outstanding loans strongly predict a nonbank’s subsequent investment. Therefore, I measure a loan’s exposure to funding from a given nonbank as the nonbank’s dollar holdings prior to the start of syndication.

Notably, my measure intentionally deviates from share-based measures of market power commonly used in the literature. Ultimately, the incremental dollar amount that an arranger would need to replace if a nonbank withheld its funds determines the cost imposed on the arranger. This cost depends on the absolute size of the funding shortfall but is only indirectly linked to the CLO manager’s relative share. Put differently, all else equal it is less costly for a bank to retain 100% of a \$1 million loan (\$1 million) than 10% of a \$100 million loan (\$10 million).

IV.1 Empirical Design

Like U.S. public securities offerings, primary market offerings in the leveraged loan market follow uniform pricing. This means that primary allocations are sold to all investors at identical primary market terms. Thus, market power effects cannot be identified from within-loan variation. Instead, I compare outcomes of loans issued by firms in the same industry in a given quarter relative to their differential exposure to a given CLO manager, after controlling for quarterly rating-implied credit risk premia. Formally, I estimate the model

$$y_l = \beta \textit{Prior Holding}_{mf(l)t(l)} + \mu_{mi(l)t(l)} + \rho_{r(l)t(l)} + \phi_{f(l)} + \kappa X_l + \varepsilon_{ml}. \quad (1)$$

This specification explains loan contract outcomes, y , for institutional term loans, l , in my sample. The variable of interest is *Prior Holding* which is my measure of CLO manager m ’s market power with respect to the loan of borrowing firm f . Specifically, this variable captures a CLO manager’s total dollar holdings in the borrower’s outstanding loans, expressed in million U.S. dollars, prior to the syndication of the new loan. I record these holdings two quarters before the new loan takes effect. This two-quarter lag ensures that holdings are measured before the start of the syndication, and, therefore, that they are not a short-term response to the syndication itself.⁸

⁸With the caveat that syndication times are available for only a small subset of my sample, the median syndication takes 56 days from the date the arranger receives the mandate to the date that the loan becomes

Manager-industry-time fixed effects, μ , ensure that I compare outcomes across loans issued by firms in the same industry i during quarter t relative to their exposure to the same CLO manager and control for managers' time-varying industry-level loan demand. Rating-quarter fixed effects, ρ , remove differences in loan outcomes due to time-varying credit risk premia at rating level, r . Firm fixed effects, ϕ , control for time-invariant borrower characteristics. Finally, I include non-price loan controls, X , listed in Table 2 and summarized in Table 1 together with other variables in my sample.⁹ Inference is based on two-way clustered standard errors that are clustered at the CLO manager level, i.e., the unit of randomization (Abadie, Athey, Imbens, and Wooldridge, 2023), and at the loan level to account for correlation in the error term induced by uniform loan pricing.

Throughout, I estimate Specification 1 using weighted estimators. I define weights to be the inverse of the number of observations per loan, which effectively equal-weights each loan. Thus, I capture the average market power effect across loans. By contrast, unweighted estimators assign equal weight to each observation and capture the average effect over CLO manager-loan pairs. This places relatively greater weight on borrowers whose loans are held by more CLO managers. If CLO funding is more substitutable for widely held loans, CLO managers' ability to exert market power could be dampened.¹⁰ As a result, the unweighted estimand may understate the average market power experienced by borrowers. Consistent with this interpretation, unweighted estimators yield around 30% smaller estimates. Nonetheless, the effect remains economically large and statistically significant. I report unweighted specifications in Internet Appendix E.

Ordinary least squares estimates of my model would recover the true parameter value of interest if there are no confounding factors that vary with a CLO manager's loan holdings within an industry that simultaneously affect loan pricing. For instance, this condition might hold in a Capital Asset Pricing Model world where an investor's holdings reflect effective.

⁹I argue that these loan terms are not jointly determined with a loan's pricing because they are typically set before and not as a result of the syndication. In Table IA.A2, I examine the effect of CLO manager market power on these terms. I find neither statistically nor economically significant effects. Often, these non-price loan terms are final before the start of underwritten syndications, which may explain the lack of detectable effects.

¹⁰I explicitly explore this hypothesis in Section V and find results consistent with increased market power for loans held by few CLO managers.

portfolio weights of the market portfolio scaled by the investor’s portfolio size. However, the assumptions needed to generate such portfolios arguably poorly approximate the leveraged loan market.

My principal concern is that a CLO manager may have private beliefs about the quality of a given borrower. These private beliefs could simultaneously affect her current holdings as well as new loan terms. For example, a CLO manager with positive beliefs about a borrower, i.e., the CLO manager believes that the expected loss of the loan under the risk-neutral measure is lower than market prices imply, would hold relatively more of that borrower’s outstanding loans. At the same time, the CLO manager would have a higher willingness to pay for the borrower’s new issue which would make the CLO manager more prone to give in to borrower-friendly loan terms.

To disentangle the effects of market power from those of private beliefs, I develop a new instrument. A growing literature uses mergers between lenders as a source of exogenous variation in exposure to lenders (e.g., [Erel, 2011](#); [Favara and Giannetti, 2017](#); [Garmaise and Moskowitz, 2006](#); [Giannetti and Saidi, 2019](#)). I extend this approach to nonbanks and utilize variation from mergers in the CLO industry. Specifically, I propose that shifts in funding exposure due to mergers between CLO managers satisfy the requirements necessary to serve as a valid instrument.

For the instrument to be valid, it must satisfy both the exclusion restriction and the relevance condition. Since an instrument’s relevance is inherently testable, I demonstrate that my instrument satisfies this condition alongside my main results. In contrast, the exclusion restriction requires a more detailed justification. The identifying assumption for the instrument to be excluded is that holdings obtained through mergers affect loan pricing only through additional nonbank market power. The main concern for this assumption is that merger-obtained holdings could also be correlated with private beliefs. This would be the case, for example, if acquirers select merger targets based on their loan portfolios. However, this concern does not reflect CLO managers’ merger decisions during the period of this study.

My argument is that acquirers pick targets based on their CLO liabilities, rather than their assets. Targets appear constrained in their CLO issuance, and their CLOs face high cost of debt. Mergers create value by refinancing these debt tranches at a lower cost. To my

knowledge, mergers in the CLO industry have not yet been studied in the literature. As a result, I am unable to point to earlier work as a reference for this claim. Instead, I provide supporting evidence as an additional contribution in this paper. However, in the interest of proceeding to the main market power results, I delegate this necessary discussion on the instrument’s exclusion restriction and related details on mergers in the CLO industry to Section VI.

To construct the instrument, I begin with my hand-collected sample of CLO management changes. The full sample consists of 84 events in which a U.S. CLO is transferred to another CLO manager. From this sample, I exclude five events in which the acquirer managed only a European CLO business prior to the transfer and four events in which all transferred U.S. CLOs were outside their reinvestment periods. These mergers do not generate changes or only limited changes in portfolios necessary for identification. Next, I remove 15 events where the seller continues to manage active U.S. CLOs after the transaction. This restriction permits partial sales where the target retains a European CLO business or CLOs past their reinvestment period. The former case is valid due to the segmentation of the U.S. and European CLO business. The latter case allows acquirers to decline the assumption of CLOs that currently wind down. However, this restriction excludes other non-merger acquisitions to avoid concerns over selection from the target’s CLOs. As an additional precaution, I exclude four mergers for which the acquirer makes reference to the target portfolio fit in public merger statements. I discuss these mergers in Section VI. The remaining 57 mergers account for 318 U.S. CLO management contracts transferred to 40 unique CLO managers.¹¹

For a given CLO manager-loan issue, the instrument equals the CLO manager’s total volume of the issuing firm’s outstanding loans obtained through the CLO manager’s prior merger activity. Merger-implied portfolio additions are measured in the quarter preceding a merger. To account only for direct portfolio effects, I discard the amount corresponding to loans matured or refinanced in the interim between merger and new loan issuance. If a CLO manager obtained multiple loans of an issuer, either because a target’s portfolio including multiple loans or through repeated merger activities, I take the total amount. Absent any applicable holdings, I set the instrument to zero.

¹¹Those numbers are comparable with studies employing bank mergers as an instrument. For example, 80 out of 210 banks in the 1992 to 1999 sample of [Garmaise and Moskowitz \(2006\)](#) were involved in mergers and [Saidi and Streit \(2021\)](#) record 79 merger events between banks for their 1990 to 2015 sample.

IV.2 Main Results

Table 2 presents estimates of Specification 1 for different pricing outcomes. The most salient contract term of leveraged loans is their credit spread. Column 1 takes the credit spread as the outcome variable. The coefficient estimate on Prior Holding is positive and statistically different from zero. This result supports the hypothesis that CLO managers exert market power to command higher spreads. Economically, the effect is large. The point estimate of 0.849 implies that a one-standard-deviation increase in my market power measure raises spreads by 13.3 bps on average. The first stage shows that a CLO manager’s merger activity significantly increases borrowers’ funding exposure to that manager and formally confirms the instrument’s relevance.

Next, I investigate how CLO manager market power affects other loan pricing outcomes. Column 2 presents estimates for the original issue discount (OID). The OID represents a loan’s primary market price as discount to par. I find that a one-standard-deviation increase in market power leads to an average increase in the OID of 6.5 bps, which means that nonbanks obtain primary allocations at a lower price. All else equal, this implies a higher expected return for primary market participants.

Similar to [Berg, Saunders, and Steffen \(2016\)](#), the effective yield combines the credit spread and OID to measure a loan’s total borrowing cost. The extent to which the OID increases a loan’s yield depends on the loan’s expected time to repayment. Leveraged loans near-universally grant the borrower the option to prepay at any date. Typically, firms exercise this option well before maturity. A common assumption is that firms call loans three years after origination, which contrasts with a median original time-to-maturity of more than six years in my sample. Under this assumption, a simple measure of a loan’s effective yield evenly distributes the loan’s OID over the three years and adds this component to the spread. The estimate in Column 3 implies that a one-standard-deviation increase in market power raises the effective yield by 15.5 bps. This magnitude closely mirrors the separate credit spread and OID estimates.

Column 4 investigates the secondary market response to market power through the effect on loan underpricing. Underpricing refers to the return from selling a primary allocation at the “break price,” which is the price of a loan’s first secondary market trade. I calculate the

break price from CLO transactions within six months of origination as the price of the first sale to an unaffiliated party.¹² I find that a one-standard-deviation increase in market power leads to 13.3 bps higher loan underpricing, which represents a large 24.8% increase over the average. Thus, CLO managers have market power to lower the price at which they obtain primary allocations relative to subsequent secondary market values.

My underpricing results provide a market-based measure of the economic costs that nonbank market power imposes on borrowers. Moreover, this finding speaks to the validity of my identification strategy. Due to uniform pricing, identification must come from outcome comparisons across loans. Thus, it is conceivable that I observe higher borrowing costs not because of market power, but rather as compensation for some unaccounted risk factor that correlates with my instrumented market power measure. However, if my primary market results were indeed the result of priced risk, this risk would also be reflected in the pricing of arm’s-length secondary market transactions. The fact that, instead, I find an effect on underpricing commensurate to that on primary market pricing rules out alternative risk-based explanations of my findings.

Market power effects are also present in other loan price outcomes. To reduce the value of a loan’s prepayment option, lenders may equip a loan with a soft call provision. Unlike hard call provisions present in corporate bonds, which preclude borrowers from calling a bond before some cutoff date, soft call provisions still permit the borrower to call the loan at any date. However, to call a loan before the end of its soft call period, the borrower must pay a premium over par. Table IA.A1 in Internet Appendix A shows that the soft call premium rises and the soft call period lengthens with CLO manager market power. This effect decreases the value of the borrower’s prepayment option and thereby increases the

¹²In general, CLOs must obtain primary allocations through secondary market purchases to maintain their pass-through vehicle status. These purchases convey no open market information, as they occur at pre-agreed primary terms. Nonetheless, they may appear as trades in CLO trustee reports. To exclude these de facto primary market trades, I remove any trades that are marked as purchases. Further, I require that the loan is sold to an unaffiliated party because trades between CLOs managed by the same manager may not resemble arm’s length transactions. Since CLOs do not disclose counterparties, I exclude trades where another CLO of the same manager purchases the loan within six months of the sale. Finally, to limit the influence of other post-primary market events on my results, I remove trades that occur more than six months after the loan becomes effective. The median break price in my sample is recorded after 45 days, and the distribution of break prices closely matches that of [Bruche et al. \(2020\)](#).

loan’s value to the lenders.¹³

IV.3 Evidence from Other Debt Markets

My empirical design requires that CLO manager mergers affect loan pricing only through CLO managers’ market power. To test this assumption, I examine whether increased exposure to a CLO manager affects borrowing costs in debt markets in which CLOs do not provide primary market funding. Since CLO managers do not participate as lenders, any effect of my instrument on pricing would suggest that my instrument captures something other than CLO managers’ capacity to influence prices.

The high-yield bond market is a natural market to start. Like institutional loans, high-yield bonds are term debt issued by sub-investment grade corporate borrowers, typically lack maintenance covenants, have minimal amortization, are widely held by institutional investors, and trade in similarly liquid secondary markets. However, unlike institutional loans, CLOs generally do not participate in the primary market for bonds.

For borrowers in my sample, I obtain high-yield bond issuance and characteristics data from LCD. I restrict this sample to plain vanilla fixed-rate coupon bonds. As in the main analysis, I define a CLO manager’s prior holding as the dollar holdings of the borrower’s outstanding loans two quarters before the bond’s issuance, and construct the instrument analogously. While all loans in my sample are secured, my bond sample includes both secured and unsecured bonds. To control for pricing differences between secured and unsecured bonds, I include an indicator that identifies unsecured bonds to the set of controls used for loans.

For comparability, I compute floating-rate equivalent spreads by deducting the maturity-matched swap rate from the bond coupon. I calculate the no-arbitrage par swap rate from the daily zero-coupon yield curve data provided by [Liu and Wu \(2021\)](#). As for institutional loans, the OID is the issuance price expressed as discount to par. The effective yield adds the OID evenly distributed over five years to the spread. The five-year effective maturity assumption reflects that bonds are often called two to three years before their stated maturity,

¹³I find no significant effect on the base rate floor, which is consistent with this being a highly standardized contract term. While floors vary with interest rates over time, there is little within-quarter variation. In most quarters, the most frequent floor level appears in more than half, and often more than 75%, of all loans.

which is slightly above eight years for the median bond in my sample. Underpricing is the return from the primary market price to the break price.

Table 3 Panel A presents the results. As in my main analysis, the first-stage estimates confirm instrument relevance across specifications. I find no statistically significant evidence of CLO manager market power affecting bond pricing in any specification. In fact, the signs of the point estimates flip across outcomes.¹⁴ Table IA.A3 confirms this finding for outcomes that do not require fixed-to-floating swaps. Again, I find no evidence of any effect on the yield-to-maturity, the T-spread, or the coupon.

Next, I investigate whether my measure of nonbank market power is associated with increased borrowing costs of pro rata tranches. In the leveraged loan market, the term pro rata tranche refers to amortizing term loans, typically called term loan A, or revolving credit lines.¹⁵ Different from bonds, these loans are covered under the same credit agreement as institutional loans, and both pro rata and institutional tranches generally rank pari-passu in bankruptcy. However, like bonds, pro rata tranches are generally not marketed to institutional investors. Instead, pro rata tranches remain with banks, which makes them a natural placebo within the leveraged loan market.

In Loan Connector, I classify loans marked as credit lines, “Term Loan A,” and non-institutional “Term Loan” as pro rata tranches. To form the sample and define the variables of interest, I follow the procedure previously described for institutional loans and high-yield bonds. I adjust the loan controls and rating-time fixed effects to allow for heterogeneity by pro rata loan type.

Table 3 Panel B reports the estimates with the spread as the primary pricing outcome. Since OIDs are broadly absent for pro rata tranches, effective yield estimates are nearly unchanged. In neither the full pro rata sample nor the credit line or the term loan subsamples do I find a statistically significant effect of CLO manager market power on the spread. For all three samples, the coefficient estimate of interest is negative. In addition to the spread, I also examine whether there is an effect on commitment fees. Borrowers pay commitment fees on the undrawn portion of their credit lines. Here, too, I find no evidence of any effect.

¹⁴Today, CLOs can have limited bond holdings. During the Volcker rule period, however, CLOs were legally precluded from holding bonds entirely. Table IA.A4 confirms the results for this subsample.

¹⁵Historically, allocations in these loans were assigned proportionally, which explains the “pro rata” label.

Frequently, borrowers issue institutional and pro rata tranches under the same credit agreement at the same time. Since both tranches are paid from the same cash flows and rank *pari passu* in bankruptcy, any differential effect of CLO manager market power on the institutional tranche but not the pro rata tranche would further point to a nonbank market power channel. I pool institutional and pro rata observations that occur in the same deal and, in the spirit of [Ivashina and Sun \(2011\)](#), estimate

$$y_l = \beta \text{Prior Holding}_{mf(l)t(l)} \times \text{Institutional}_l + \delta_{mf(l)t(l)} + \rho_{r(l)\tau(l)t(l)} + \kappa_{\tau(l)} X_l + \varepsilon_{ml}. \quad (2)$$

This specification extends Specification 1 by interacting my main measure with the indicator *Institutional* that identifies institutional tranches. The coefficient β captures the differential effect of CLO manager market power on institutional relative to pro rata tranches. When possible, I also include the main effect of the uninteracted measure. I allow for loan type heterogeneity in rating controls by including rating-loan type-quarter-fixed effects, ρ , and for loan type heterogeneity in loan control slopes. The loan type (τ) categorizes a loan as institutional loan, bank term loan, or credit lines. Lastly, and importantly, I include manager-firm-time fixed effects δ . Because each deal in my sample is uniquely identified by a firm-quarter, δ corresponds to manager-deal fixed effects. These fixed effects absorb any deal-level characteristics common across all tranches in the same deal such as borrower credit risk or loan demand ([Khwaja and Mian, 2008](#)).

Table 4 presents the results with the effective yield as the outcome variable. Column 1 measures market power effects for institutional and pro rata loans separately. As in my earlier placebo tests, I do not find any statistically or economically meaningful effect for pro rata tranches. In contrast, for institutional loans the interaction term is positive and statistically significant, with a point estimate of 0.763. This effect on loans that are issued jointly with pro rata tranches is only slightly weaker than the effect for the total loan sample. In Column 2, I replace the pro rata tranche interaction term with the prior holding main effect. This specification confirms that the effect on institutional loans is not only larger than zero, but also statistically distinguishable from the effect on the pro rata tranches. Lastly, I include manager-deal fixed effects in Column 3. The resulting coefficient estimate remains nearly unchanged at 0.710. While the estimate becomes noisier, it remains statistically significant at the 10% level. Hence, I identify an effect that exists only for tranches where

CLO managers provide primary market funding. Overall, this provides strong evidence in support of my market power interpretation.

V Sources of Market Power

I have demonstrated that CLO managers possess market power in leveraged loan origination. The ability of CLO managers to influence loan pricing in their favor indicates that arrangers cannot costlessly replace a CLO manager’s funds with funds from other sources. In this section, I investigate the information environment and constraints in funding alternatives as sources of this limited substitutability.

V.1 The Information Environment

Information asymmetries are central to the pricing of traditional bank loans. In contrast to public market investors, banks incorporate private information, which they collect on their borrowers, in their lending decisions. In seminal contributions, [Sharpe \(1990\)](#) and [Rajan \(1992\)](#) recognize that a bank’s possession of private information subjects its uninformed competitors to adverse selection. This creates an information monopoly that allows the incumbent to lend at a lower rate than its competitors but above its cost of funds.

Borrowers and arrangers may face similar information frictions with respect to nonbank lenders in the leveraged loan market. If nonbank lenders are asymmetrically informed about a borrower, funds from uninformed nonbanks to replace an incumbent may require additional compensation for adverse selection. Consequently, invested nonbanks may enjoy an “information oligopoly.”¹⁶

If market power arises because of information asymmetries that make a lender costly to replace, the pricing effect should be stronger in less transparent information environments and weaker when outside investors can more easily evaluate the borrower. Table 5 tests this hypothesis. Across specifications, I focus on the effective yield which summarizes the credit spread and original issue discount into a single measure of overall borrowing costs. The specification extends my baseline Specification 1 and includes the interaction of a manager’s

¹⁶Alternatively, the term “information oligopsony” could highlight that nonbank lenders purchase loans.

prior holdings with indicators that capture differences in the quality of information available about the borrower.¹⁷ For this variable, I include the interaction of the instrument with the indicator as a second excluded instrument.

In Column 1 and Column 2, the indicators *Sponsored* and *Public* identify private equity sponsored and publicly-listed borrowers. Sponsors reduce information gaps through certification and monitoring (e.g., [Haque, Mayer, and Wang, 2025](#)). Public firms directly reduce information asymmetries through extensive public disclosures. The negative coefficient estimates for both imply that loans of sponsored and public borrowers are subject to less market power. For sponsored borrowers, the effect is reduced by about a third relative to non-sponsored firms, with a p -value of 10.7%. The coefficient on public is highly statistically significant. Its magnitude suggests that market power is concentrated in private firms.

Next, I study heterogeneity with respect to firm size as smaller borrowers tend to be harder to evaluate. An empirical challenge is that private firms make little information about their size publicly available. To still measure this dimension for private firms, I obtain private firm revenue data from PitchBook. For public firms, I obtain revenue data from Compustat. In total, private firms represent nearly half of the non-missing revenue sample. I classify firms in the bottom revenue quartile as small. The positive and significant coefficient estimate on the interaction term in Column 3 indicates increased vulnerability to nonbank market power for smaller firms.

Column 4 assesses heterogeneity with respect to available secondary market information. *Illiquid* equals one if the number of trades of a borrower’s outstanding loans two quarters before syndication is in the bottom quartile relative to other issuers in that quarter. For borrowers with illiquid loans, secondary market prices convey less information about a borrower’s credit quality. I find that CLO managers command more favorable terms when a borrower’s outstanding loans are illiquid. The size of the coefficient is large. Though noisier than the other estimates, the estimate remains significant at the 5% level.

Another measure common in the corporate credit literature is the presence of a public credit rating. In my sample, all loans are rated because CLOs are generally restricted from investing in unrated loans. Thus, I cannot use such an indicator of information availability. Instead,

¹⁷When not subsumed by fixed effects, the indicator is included with the vector of controls.

I use the institutional feature that most CLOs require ratings from multiple agencies for a loan to be an eligible investment. The presence of multiple credit ratings allows me to capture rating disagreement. In the top quartile of disagreement, ratings disagree by more than one rating notch. I let *Rating Uncertainty* take the value one for these borrowers. The rationale for this measure is that public ratings serve to communicate private information about borrowers to public-side investors. Disagreeing ratings provide a less informative signal, which leaves greater information asymmetries among investors. Column 5 shows that rating disagreements substantially increase nonbank market power.

Overall, the results of Table 5 support an information-based explanation of nonbank market power. Consistent with the idea that CLO manager market power is reduced when information about the borrower is more readily available to other investors, I find weaker effects for sponsored and public borrowers. In contrast, the effect is stronger for smaller borrowers, borrowers with illiquid outstanding loans, and borrowers with credit rating disagreements. These are precisely the settings in which outside investors are likely to face greater difficulty assessing credit risks, which gives leverage to incumbent informed investors. Thus, these results strengthen the paper’s mechanism by establishing that prior holdings matter the most when they proxy for hard-to-replace informed capital.

V.2 Funding Availability

Information frictions reduce the willingness of uninformed lenders to supply additional funds. Table 6 studies how overall capital availability shapes nonbank market power. These tests complement the information-based results by focusing on the supply of outside capital rather than on borrower transparency. If market power reflects the difficulty of raising additional nonbank funds, the effect of prior holdings should be strongest when nonbank funding supply is constrained. I interact my main measure with indicators for the share of CLOs past their reinvestment period, concentration of the existing lender base, whether the loan finances an acquisition, and borrower access to the bond market.

Across specifications, Table 6 clearly shows that capital scarcity increases CLO managers’ ability to exert market power. CLOs are finite life investment funds. As a CLO approaches maturity, its reinvestment period ends at which point the CLO can no longer reinvest proceeds into new loans. The indicator in Column 1 identifies quarters in the bottom quartile

by the share of CLOs in their reinvestment period. As fewer CLOs remain able to reinvest, capital becomes less available. The positive coefficient on the interaction term means that these periods are associated with increased market power effects. While fewer reinvesting CLOs generally reduce capital availability, a thin and highly concentrated existing lender base makes it more difficult to raise additional funds from existing nonbank lenders. The indicator in Column 2 equals one for loans in the top quartile by the Herfindahl-Hirschman Index of CLO manager holdings within their quarter. The evidence indicates that nonbank lender concentration amplifies market power.

Column 3 compares market power for loans in acquisition deals to non-acquisition deals. In acquisition deals, borrowers typically have less flexibility to reduce the loan size. The coefficient is again positive, though noisier, consistent with increased market power when the borrower cannot easily reduce the loan amount. Lastly, the effect is also stronger for borrowers without bond market access.

Together, Table 5 and Table 6 move beyond documenting average effects and identify the settings in which CLO manager market power matters most. Table 5 finds that the effect is stronger where information about borrowers is scarce, while Table 6 shows that it is stronger when the availability of outside funding options is limited. This heterogeneity is important as it ties the main pricing results to an underlying mechanism. CLO managers can extract more favorable terms when information about a borrower is less transparent to outside investors and when capital is scarce. These results help further distinguish the mechanism from generic demand or risk effects and add a more precise economic interpretation to the broader findings.

VI Mergers in the CLO Industry

In the preceding analysis, I have posited that variation from mergers between CLO managers provides a valid instrument to identify nonbank market power in the origination of leveraged loans. In Section IV, I established the instrument’s relevance. To justify the instrument’s exclusion restriction, I study the determinants of mergers in the CLO industry in this section.

Recall, for the instrument to satisfy the exclusion restriction, I require that merger-induced changes in a loan’s funding exposure to a CLO manager are uncorrelated with the manager’s

private beliefs about borrower quality. The main concern is that such correlation could arise if acquirers' private beliefs also affect merger target selection. However, I find that portfolios do not drive target selection. Instead, acquirers select targets that face constraints in their ability to issue CLOs.

I provide several pieces of evidence that, taken together, support my identifying assumption. First, in the time-series, high M&A activity coincides with difficult CLO issuance conditions. Second, almost 90% of public merger statements cite improved access to capital as a motivation for the merger, while only four statements contain any reference to the target's loan portfolio.¹⁸ Third, I provide direct evidence that targets are indistinguishable from other similarly-sized managers based on their portfolio composition.

VI.1 The History of Mergers in the CLO Industry

Figure 3 compares the annual number of U.S. CLOs transferred to another manager against new CLO issuance volumes. Following the Global Financial Crisis, issuance conditions were unfavorable to new CLOs. Issuance halted for nearly two years during which the CLO industry consolidated. It took until the end of 2010 for the CLO issuance market to restart with a new generation of CLO structures known as CLO 2.0. As issuance picked up, M&A activity subsided, which led to only a single merger in 2013 and 2014.

In late 2014, U.S. regulators ruled that CLOs were subject to risk retention requirements under the Dodd-Frank Act. These "skin-in-the-game" provisions stipulated that CLO managers must retain at least five percent of a CLO's value. In response, M&A activity resurged as many CLO managers sought to either improve their institutional capital access or to exit the market. Concurrently, issuance fell as CLO managers built up capital in anticipation of the rule's implementation.

Risk retention requirements came into effect at the end of 2016. However, ultimately, these rules remained short-lived. In 2018, a lawsuit won by the LSTA reversed their application to CLOs which was followed by high CLO issuance activity. Issuance declined again in mid-2019 and continued to fall in 2020 due to the COVID-19 pandemic. During this time M&A

¹⁸Although these references are general in nature, I have excluded these mergers for the construction of the instrument.

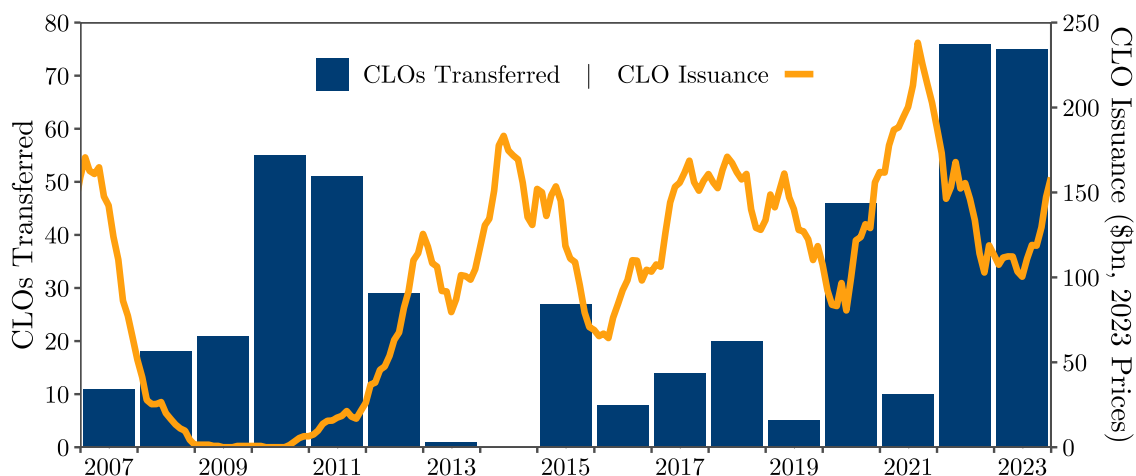


Figure 3. M&A Activity in the U.S. CLO Industry

This figure depicts the U.S. CLO merger and issuance activity between 2007 and 2023. Bars describe the annual number of U.S. CLOs that are transferred to another U.S. CLO manager, with values on the left axis. The line shows the annualized volume of CLOs issued over a six-month centered rolling window in billion real U.S. dollars at 2023 prices, with values on the right axis.

activity approached levels seen last during the Global Financial Crisis.

In 2021, merger activity was low as the CLO market recorded all-time-high issuance volumes. This brief recovery period ended with the onset of the recent monetary tightening cycle. Since then, CLO issuance has been low while M&A activity has reached record levels.

In summary, merger activity in the CLO industry responds countercyclically to CLO issuance which points to CLO funding conditions as a key determinant of mergers.

VI.2 Merger Rationales from Public Statements

Next, I employ a narrative approach to study what motivates CLO manager mergers. For the mergers in my sample, I collect public statements that accompany these transactions. I am able to find statements for 48 mergers. In the 13 other cases, the parties did not comment publicly.¹⁹ For each public statement, I identify comments on the motivation for the merger. I classify these comments into four broad categories: capital access, business operations,

¹⁹These mergers tend to involve small targets in the early sample period. In four cases, the involved parties explicitly declined to comment. For the other cases, references appear only in rating agency statements or collateral replacement agreements.

non-U.S. CLO markets, and loan portfolio consideration. I further remove two mergers for which only unspecific comments are available such as on positive return expectations for investors.

Figure 4 shows the share of mergers that reference each comment and category. The most frequently cited category for merging is improved capital access. Reasons in this category are mentioned for nearly nine out of ten mergers. Within this category, CLO managers frequently cite availability of both equity and outside investor capital. The combined CLO manager benefits from improvements in investor relationships. Half of all mergers claim improved capital access as the result of additional scale. For example, in reference to their acquisition of Gulf Stream Asset Management, a representative of Apollo Global Management stated, “[...] while smaller market participants can continue to add a high level of service to their existing investors, without the scale and global reach of a platform such as Apollo’s, the smaller firms are unable to effectively compete in the market to raise new funds, and their growth prospects are limited.” Operational reasons, such as cost synergies, occur for 48% of mergers. Benefits for activities in other markets, such as the European CLO market or non-CLO markets like private credit or equities, are mentioned in 38% of cases.

In contrast, the least frequent category contains comments about loan portfolios. Only four statements make any reference to the loan assets, all general statements about the fit of the target’s assets. To illustrate, one acquirer noted that the target’s CLOs “[...] have been well managed in a manner consistent with our style and philosophy such that the assets are very complimentary [sic] with our existing portfolios.” In another case, the acquirer claimed that the “assets are complementary to our existing CLO portfolios.” None of these statements explain the source of complementarity, and all but one prominently cite other considerations as well. Therefore, it is unclear whether these mergers would pose a problem to the instrument’s validity. Nonetheless, I have excluded these mergers in the construction of my instrument.²⁰

²⁰When I include these mergers, I obtain slightly smaller estimates but the statistical or economic significance of my results remains unchanged.

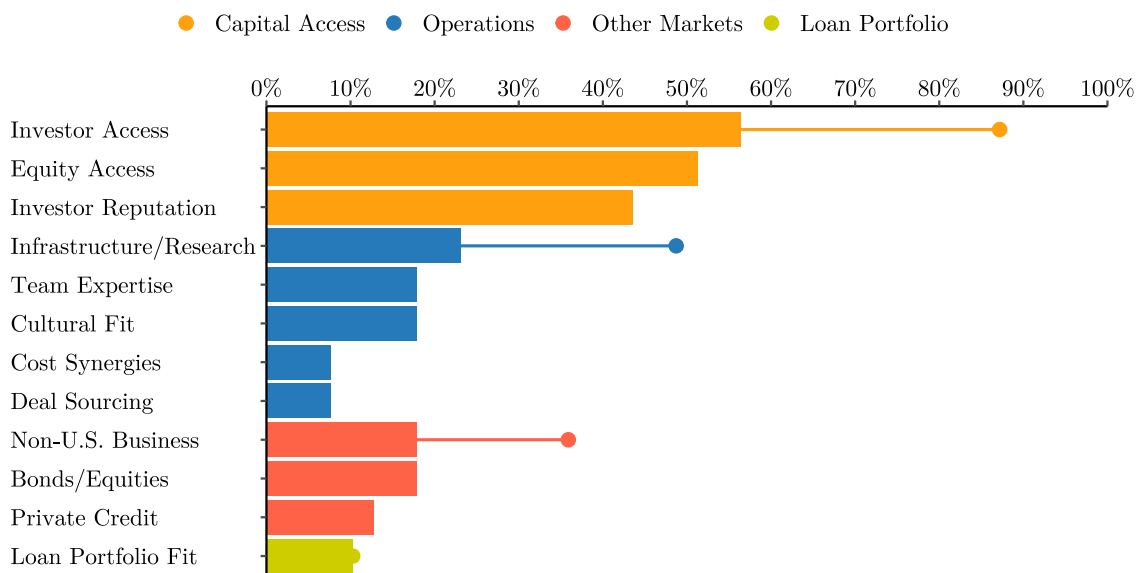


Figure 4. Stated Merger Rationales

This plot describes publicly stated motivations for mergers in the U.S. CLO industry. Motivations are classified into twelve categories in one of four groups. The percentage share of mergers that mention a given category is displayed by colored bars. The color of a given bar indicates the category's group. Points display the share of mergers that mention any category within of a group.

VI.3 Testing Portfolio-Based Target Selection

So far, the presented evidence has shown that M&A activity in the CLO industry behaves countercyclically to CLO issuance and that improved capital access is a main rationale for merging. I now proceed to examine the role of loan portfolios in merger target selection directly. I test if their loan portfolio distinguishes the target from other CLO managers in this set, conditional on the loan portfolio of the acquirer. For this exercise, I define a set of counterfactual similar-sized target candidate CLO managers. For each merger, I include a manager in this set if they manage between half and double the number of CLOs as the target at merger announcement. Notably, the unobserved actual candidates that an acquirer had considered would not be the correct counterfactual group because these CLO managers may already have been selected for their loan portfolios.

In Table 7, I present statistics for the acquirer, the target, and the target candidates. By construction, target and target candidates have a similar size when measured by the number of CLOs under management or the number of loans in their portfolio. Furthermore, targets

and target candidates are comparable by portfolio metrics that I did not select on. Importantly for my instrument, target and target candidates' portfolios appear on average similar in their portfolio alignment with the acquirer in terms of overlap and portfolio similarity as measured by the cosine similarity.

To formally test whether the target's portfolio alignment with the acquirer statistically separates the target, and to control for other factors that potentially could mask an effect, I estimate the linear probability model

$$Target_{am} = \beta Portfolio\ Alignment_{am} + \alpha_a + \kappa X_{mt(a)} + \varepsilon_{am}. \quad (3)$$

The dependent variable, *Target*, is a dummy variable that equals one for the realized target manager, m , in acquisition, a , and zero for non-selected target candidates. I consider two measures of portfolio alignment with the acquirer. First, *Portfolio Overlap* measures the value-weighted share of loans that a target candidate portfolio has in common with the acquirer. Second, *Portfolio Similarity* measures the cosine similarity between portfolio weight vectors of the target candidate and of the acquirer. This metric captures differences in portfolio tilts towards the acquirer at the intensive margin. As controls, I include merger fixed effects, α , to absorb factors common to all target candidates such as the acquirer's portfolio specificity relative to all target candidates. Lastly, I include a set of control variables, X , listed in Table 8.

Table 8 Panel A displays estimates of Specification 3. Column 1 and Column 2 show that neither portfolio overlap nor portfolio similarity can statistically distinguish the target. The coefficients are positive for the portfolio overlap and negative for the portfolio similarity. Statistically, the coefficient estimates are indistinguishable from zero in all specifications. The implied size of the economic effect is small. For portfolio overlap, a target candidate with an overlap equal to the 75th percentile of the distribution is only 0.25 percentage points more likely to be the realized target than an otherwise equal CLO manager with an overlap equal to the 25th percentile. This effect is even smaller for the portfolio similarity. Column 3 and Column 4 find similarly small magnitudes for portfolio alignments measured at the loan issuer level.

Potentially, manager-level portfolio measures are too coarse to distinguish the targets. To

address this concern, I use the granularity of my data and expand my sample to the merger-CLO manager-loan/issuer level. I estimate the linear probability model

$$Target_{am} = \beta Portfolio\ Alignment_{aml} + \alpha_{al} + \kappa X_{mt(a)} + \varepsilon_{aml}, \quad (4)$$

on the sample of issuers or loans in the acquirer portfolio. I again consider two measures of portfolio alignment. The indicator Shared Issuer/Loan equals one for loans held by both target candidate and acquirer. The second measure is the negative absolute difference between the target candidate’s portfolio weight, w , and the acquirer’s portfolio weight, w^A . For easier comparability with earlier tests, I take the negative so that an increase signifies closer portfolio alignment with the acquirer. I now include merger-loan fixed effect, α_{al} to average out acquirer-specific portfolio idiosyncrasies at the loan level. This set of fixed effects also controls for any loan characteristic at time of merger announcement. Finally, I include the same vector of CLO manager characteristics as in previous manager-level tests.

Panel B in Table 8 tabulates the resulting OLS estimates. Despite the additional granularity of these tests, target loan holdings continue to be indistinguishable from those of other CLO managers. Again, this holds true for both portfolios considered at the issuer level in Columns 1 and 2, and the loan level in Columns 3 and 4. The absence of an effect is independent of the sample definition in this specification.

Overall, loan portfolios do not appear to be prominent determinants of mergers. Perhaps, this is not surprising. To prevent risk shifting, CLO noteholders require that a CLO must maintain high levels of diversification through covenants that tightly restrict portfolio concentration (Kundu, 2022). Typical concentration limits dictate individual borrower exposures below 2%, industry exposures below 15%, and CCC-rated high credit risk exposures below 7.5%. Additional requirements like industry diversity score tests further constrain possible concentration. At the manager level, portfolio diversification is even higher because individual CLO portfolios are not perfectly correlated.

VII Conclusion

This paper establishes that nonbank lenders materially affect the terms at which firms borrow in the leveraged loan market. My central finding is that CLO managers possess market power in primary loan syndication. CLO managers who impose greater funding risk are able to command higher spreads and purchase loans at larger discounts. Higher underpricing in the secondary market and the absence of an effect in placebo markets confirm that increased borrowing costs do not compensate for priced risks. The effect on firms' cost of borrowing is economically important. These patterns imply that competition in the modern loan market is shaped not only by banks, but also by the structure of nonbank demand.

The paper identifies why this market power arises. Investor-specific information and limited available funding alternatives make a manager's funds harder to replace. The effect of CLO manager market power is stronger for opaque borrowers and weaker where information is widely available. It is also stronger when holdings of a borrower's loans are concentrated, and when the borrower lacks access to the bond market. These findings point to a setting in which investor participation is valuable because other investors cannot be substituted in at low cost. Thus, this paper extends the logic of informational rents beyond traditional bank relationships and into nonbank intermediaries in underwrite-to-distribute market structures.

A further contribution of this paper is to empirically isolate nonbank market power effects. My measure of CLO manager holdings captures both a loan's funding exposure to the manager and the manager's revealed demand shaped by private beliefs about the borrower. To separate the market power effect, the paper develops an instrument based on mergers in the CLO industry. I argue that merger-induced shifts in exposure serve as a valid instrument because loan portfolios are not determinants of merger selection. The paper's merger analysis supports this argument and contributes evidence of CLO-liability-driven mergers.

More broadly, the paper documents that the organization of nonbank intermediation matters for corporate financing outcomes. The cost of debt depends not only on borrower risk and aggregate credit conditions, but also on who provides capital and how easily that capital can be replaced. As credit provision continues to migrate away from bank balance sheets and toward managed investment vehicles, understanding the industrial organization of these institutional investors is becoming increasingly important.

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Tables

Table 1. Summary Statistics

This table reports descriptive statistics for variables used in the main analysis. The sample consists of institutional loans originated between January 2010 and December 2022 that have recorded primary allocations to CLOs and are issued by borrowers with outstanding loans held by CLOs. Panel A reports loan-level variables, Panel B deal-level variables, and Panel C loan-CLO manager-level variables.

Variable	Unit	Mean	Std. Dev.	p25	p50	p75	Obs.
Panel A: Loan-Level							
Spread	bps	405.91	168.40	300.00	375.00	450.00	3,800
Original Issue Discount	bps	73.90	108.93	0.00	50.00	100.00	3,800
Effective Yield	bps	430.54	188.74	307.29	383.33	491.67	3,800
Underpricing	bps	53.55	107.47	12.56	50.25	100.25	2,434
Amount	\$mn	782.60	783.24	300.00	525.00	996.00	3,800
Maturity	mos	73.53	14.39	65.00	77.00	84.00	3,800
Cov-lite	0/1	0.72	0.45	0.00	1.00	1.00	3,800
First Lien	0/1	0.92	0.27	1.00	1.00	1.00	3,800
Panel B: Deal-Level							
Rating		13.51	1.43	13.00	13.00	13.00	3,265
Refinancing	0/1	0.80	0.40	1.00	1.00	1.00	3,265
Sponsored	0/1	0.64	0.48	0.00	1.00	1.00	3,265
Public	0/1	0.34	0.47	0.00	0.00	1.00	3,265
Small	0/1	0.25	0.43	0.00	0.00	0.00	1,669
Illiquid	0/1	0.28	0.45	0.00	0.00	1.00	3,265
Rating Uncertainty	0/1	0.16	0.37	0.00	0.00	0.00	3,030
Few Reinvesting CLOs	0/1	0.22	0.42	0.00	0.00	0.00	3,265
Concentrated Holdings	0/1	0.26	0.44	0.00	0.00	1.00	3,265
Acquisition	0/1	0.23	0.42	0.00	0.00	0.00	3,265
No Bond Market Access	0/1	0.55	0.50	0.00	1.00	1.00	3,265
Panel C: Loan-CLO Manager-Level							
Prior Holding	\$mn	11.53	15.70	2.92	6.05	13.80	112,341

Table 2. CLO Manager Market Power and Loan Pricing

This table examines the effect of CLO manager market power on loan pricing and displays estimates of

$$y_l = \beta \text{Prior Holding}_{mft} + \mu_{mit} + \rho_{rt} + \phi_f + \kappa X_l + \varepsilon_{ml}.$$

The unit of observation is a CLO manager-loan pair (ml). Prior Holding measures the CLO manager's holdings of the borrower's loans two quarters before loan l takes effect. The specification includes manager-industry-quarter (μ), rating-quarter (ρ), and borrower (ϕ) fixed effects (FE), and a loan control vector (X) containing Ln Amount, Ln Maturity, Cov-lite, First Lien, Sponsored, Refinancing, and purpose fixed effects. Estimates are obtained via two-stage weighted least squares (2SWLS), with Prior Holding (Merger) as excluded instrument for Prior Holding and each observation weighted by the inverse of the number of observations for the corresponding loan. First-stage estimates and weak instrument test statistics are reported below each specification. Standard errors, in parentheses below their point estimates, are double-clustered at the CLO manager- and loan-levels. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels.

	Spread (1)	OID (2)	Effective Yield (3)	Underpricing (4)
Prior Holding	0.849*** (0.120)	0.414*** (0.132)	0.987*** (0.153)	0.846*** (0.117)
Estimator	2SWLS	2SWLS	2SWLS	2SWLS
Manager-Industry-Quarter FE	Yes	Yes	Yes	Yes
Rating-Quarter FE	Yes	Yes	Yes	Yes
Borrower FE	Yes	Yes	Yes	Yes
Loan Controls	Yes	Yes	Yes	Yes
Observations	106,061	106,061	106,061	76,815
<i>First-Stage Estimates</i>				
Prior Holding (Merger)	0.458*** (0.045)	0.458*** (0.045)	0.458*** (0.045)	0.469*** (0.038)
Cragg-Donald F -Statistic	933	933	933	485

Table 3. CLO Manager Market Power and Loan Pricing: Placebos

This table examines whether CLO manager market power affects pricing in debt markets in which CLO managers do not provide primary market funding. The table displays estimates of Specification 1. For borrowers in the Table 2 sample, Panel A and Panel B examine high-yield bond and pro rata loan issues. Bond controls add the indicator Secured to the loan controls. Sample, variable construction, estimation, and inference are as described in Table 2. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels.

Panel A. High-Yield Bonds

	Spread (1)	OID (2)	Effective Yield (3)	Underpricing (4)
Prior Holding	0.117 (0.187)	-0.287 (0.264)	0.059 (0.198)	-0.206 (0.211)
Estimator	2SWLS	2SWLS	2SWLS	2SWLS
Manager-Industry-Quarter FE	Yes	Yes	Yes	Yes
Rating-Quarter FE	Yes	Yes	Yes	Yes
Borrower FE	Yes	Yes	Yes	Yes
Bond Controls	Yes	Yes	Yes	Yes
Observations	49,000	49,000	49,000	23,596
<i>First-Stage Estimates</i>				
Prior Holding (Merger)	0.553*** (0.021)	0.553*** (0.021)	0.553*** (0.021)	0.536*** (0.034)
Cragg-Donald <i>F</i> -Statistic	469	469	469	406

Panel B. Pro Rata Tranches

	Spread			Commitment Fee
	(1)	(2)	(3)	(4)
Prior Holding	-0.235 (0.150)	-0.206 (0.250)	-0.136 (0.109)	0.031 (0.045)
Estimator	2SWLS	2SWLS	2SWLS	2SWLS
Manager-Industry-Quarter FE	Yes	Yes	Yes	Yes
Rating-Quarter-Loan Type FE	Yes	Yes	Yes	Yes
Borrower FE	Yes	Yes	Yes	Yes
Loan Controls	Yes	Yes	Yes	Yes
Sample	All PR Loans	Term Loans	Credit Lines	Credit Lines
Observations	107,914	14,079	86,526	37,862
<i>First-Stage Estimates</i>				
Prior Holding (Merger)	0.481*** (0.047)	0.423*** (0.073)	0.475*** (0.062)	0.407*** (0.042)
Cragg-Donald <i>F</i> -Statistic	740	213	471	307

Table 4. CLO Manager Market Power and Loan Pricing: Within-Deal Evidence

This table examines the differential effect of CLO manager market power on institutional relative to pro rata loans issued under the same credit agreement and displays estimates of

$$Effective\ Yield_l = \beta Prior\ Holding_{mft} \times Institutional_l + \delta_{mft} + \rho_{\tau\tau} + \kappa_{\tau} X_l + \varepsilon_{ml}.$$

A deal is defined by a firm-quarter pair. The loan sample includes pro rata and institutional loans of deals that have a pro rata loan and an institutional loan in the sample of Table 2. For each loan, the sample includes the same managers as in Table 2 for that deal. Loan Type (τ) takes the values institutional loan, bank term loan, or credit line. Indicator Pro Rata equals one for bank term loans and credit lines; indicator Institutional equals one for institutional loans. Loan controls permit heterogeneity by loan type. Estimation and inference are as described in Table 2. Table IA.F1 reports first-stage estimates. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels.

	Effective Yield		
	(1)	(2)	(3)
Prior Holding			
× Institutional	0.763*** (0.283)	0.706** (0.329)	0.710* (0.389)
× Pro Rata	0.057 (0.252)		
Prior Holding		0.057 (0.252)	
Estimator	2SWLS	2SWLS	2SWLS
Manager-Deal FE	—	—	Yes
Manager-Industry-Quarter FE	Yes	Yes	—
Rating-Quarter-Loan Type FE	Yes	Yes	Yes
Borrower FE	Yes	Yes	—
Loan Controls	Yes	Yes	Yes
Observations	116,289	116,289	116,289
<i>Weak Instrument Tests</i>			
Cragg-Donald <i>F</i> -Statistic	426	426	852

Table 5. CLO Manager Market Power and the Information Environment

This table examines the effect of a borrower's information environment on CLO manager market power and displays estimates of

$$Effective\ Yield_l = \beta_1 Prior\ Holding_{mft} \times Indicator_{ft} + \beta_2 Prior\ Holding_{mft} + \mu_{mit} + \rho_{rt} + \phi_f + \kappa X_l + \varepsilon_{ml}.$$

Indicator stands for five different indicators: Sponsored identifies borrowers with a private equity sponsor, Public identifies publicly-listed borrowers at the start of syndication, Small identifies borrowers in the bottom quartile by revenue, Illiquid identifies borrowers whose outstanding loans are in the bottom quartile by trading number relative to other issuers in that quarter, and Rating Uncertainty identifies borrowers whose credit ratings disagree by more than one notch. When not subsumed by fixed effects, the loan control vector includes the Indicator. Sample, other variables, estimation, and inference are as described in Table 2. Table IA.F2 reports first-stage estimates. ***, **, and * denote statistical significance at the 1%-, 5%-, and 10%-levels.

	Effective Yield				
	(1)	(2)	(3)	(4)	(5)
Prior Holding					
× Sponsored	-0.402 (0.248)				
× Public		-1.434*** (0.380)			
× Small			0.600** (0.280)		
× Illiquid				1.746** (0.861)	
× Rating Uncertainty					4.019*** (1.053)
Prior Holding	1.208*** (0.207)	1.578*** (0.292)	0.843*** (0.315)	0.894*** (0.172)	0.455*** (0.122)
Estimator	2SWLS	2SWLS	2SWLS	2SWLS	2SWLS
Manager-Industry-Quarter FE	Yes	Yes	Yes	Yes	Yes
Rating-Quarter FE	Yes	Yes	Yes	Yes	Yes
Borrower FE	Yes	Yes	Yes	Yes	Yes
Loan Controls	Yes	Yes	Yes	Yes	Yes
Observations	106,061	106,061	56,897	106,061	99,624
<i>Weak Instrument Tests</i>					
Cragg-Donald <i>F</i> -Statistic	270	316	151	329	99

Table 6. CLO Manager Market Power Loan Pricing and Funding Availability

This table studies the effect of funding availability on CLO manager market power and displays estimates of

$$Effective\ Yield_l = \beta_1 Prior\ Holding_{mft} \times Indicator_{ft} + \beta_2 Prior\ Holding_{mft} + \mu_{mit} + \rho_{rt} + \phi_f + \kappa X_l + \varepsilon_{ml}.$$

Indicator stands for four different indicators: Few Reinvesting CLOs identifies quarters in the bottom quartile by the share of CLOs in their reinvestment period, Concentrated Holdings identifies the top quartile of loans in their quarter by the Herfindahl-Hirschman Index of CLO manager holdings, Acquisition identifies loans that finance an acquisition, and No Bond Market Access identifies borrowers that have not issued a public bond prior to the loan. When not subsumed by fixed effects, the loan control vector includes the Indicator. Sample, other variables, estimation, and inference are as described in Table 2. Table IA.F3 reports first-stage estimates. ***, **, and * denote statistical significance at the 1%-, 5%-, and 10%-levels.

	Effective Yield			
	(1)	(2)	(3)	(4)
Prior Holding				
× Few Reinvesting CLOs	0.489* (0.271)			
× Concentrated Holdings		2.883*** (0.374)		
× Acquisition			1.835 (1.317)	
× No Bond Market Access				1.082*** (0.373)
Prior Holding	0.766* (0.460)	0.581*** (0.166)	0.764** (0.318)	0.704*** (0.155)
Estimator	2SWLS	2SWLS	2SWLS	2SWLS
Manager-Industry-Quarter FE	Yes	Yes	Yes	Yes
Rating-Quarter FE	Yes	Yes	Yes	Yes
Borrower FE	Yes	Yes	Yes	Yes
Loan Controls	Yes	Yes	Yes	Yes
Observations	106,061	106,061	106,061	106,061
<i>Weak Instrument Tests</i>				
Cragg-Donald <i>F</i> -Statistic	251	362	192	247

Table 7. Characteristics of Merging CLO Managers.

This table compares target CLO managers to alternative target candidates. For each acquisition, the target candidates set contains CLO managers, other than the acquirer, that manage between half and double the number of CLOs as the target. The columns *Acquirer*, *Target*, *Target Candidate* display averages over the sample mergers, with the target candidate value corresponding to the median value for each merger. Variables are measured before merger announcement.

Variable	Unit	Acquirer	Target	Candidates
Panel A: Managed CLOs				
No. CLOs		14.87	7.26	7.24
No. Active CLOs		11.45	5.68	5.73
No. CLOs in Warehouse		1.61	0.21	0.60
Share Middle Market CLOs	%	13.20	9.14	9.17
Avg. Portfolio Size	\$mn	464.83	379.57	421.06
Avg. Time Since Original Close	mos	35.51	47.29	40.69
Avg. Time Since Last Issue	mos	28.24	38.14	32.33
Avg. Remaining Reinvestment Period	mos	27.79	20.28	27.17
Avg. Time To Maturity	mos	116.62	109.36	113.80
Panel B: Portfolio Characteristics				
No. Loans		450.82	343.26	345.81
No. Borrowers		342.13	265.42	269.43
Portfolio Overlap	%	100.00	35.39	35.06
Portfolio Similarity	%	100.00	12.30	14.03
Max. Loan Exposure	%	1.24	1.34	1.37
Max. Borrower Exposure	%	1.58	1.61	1.67
Weighted Avg. Spread	bps	359.75	358.95	364.56
Share IG Rated	%	1.39	1.84	1.48
Share BB Rated	%	20.13	19.28	19.25
Share B Rated	%	73.15	71.67	72.53
Share CCC Rated	%	4.15	5.55	4.91
Share Defaulted	%	1.16	1.66	1.83
Share Structured Finance	%	0.26	0.17	0.21

Table 8. CLO Manager Mergers and Portfolio-based Target Selection.

This table examines the role of loan portfolios in merger target selection and displays estimates of

$$Target_{am/aml} = \beta Portfolio\ Alignment_{am/aml} + \alpha_{a/al} + \kappa X_{mt(a)} + \varepsilon_{am/aml}.$$

The sample includes the target candidates defined in Table 7. The dependent variable, *Target*, equals 1 for the target and zero otherwise. All independent variables are measured before merger announcement. Columns 1 and 2 compare portfolios at the borrower level. Columns 3 and 4 compare portfolios at the loan level. In Panel A, the unit of observation is at the merger-manager (*am*) level. Columns one and three measure the portfolio share of loans also held by the acquirer, *Portfolio Overlap*. Columns two and four measure the cosine similarity between acquirer and target candidate portfolio weight vectors, *Portfolio Similarity*. In Panel B, the unit of observation is at the merger-manager-loan (*aml*) level. Controls include merger and merger-loan fixed effects (FE), α , and a vector of CLO manager controls, X . Controls in X are *No. CLOs*, *No. CLOs in Warehouse*, *Share Structured Finance*, *Share CCC Rated*, *Share Defaulted*, and *Share Middle Market CLOs*. Double-clustered standard errors at the merger and manager level are reported in parentheses below their corresponding OLS point estimates. ***, **, and * denote statistical significance at the 1%-, 5%-, and 10%-level, respectively.

Match Level	Target			
	Issuer		Loan	
	(1)	(2)	(3)	(4)
Panel A: Manager-Level Portfolio Alignment				
Portfolio Overlap	0.020 (0.033)		0.018 (0.033)	
Portfolio Similarity		-0.066 (0.049)		-0.082 (0.054)
Merger FE	Yes	Yes	Yes	Yes
Manager Controls	Yes	Yes	Yes	Yes
Observations	1,848	1,848	1,848	1,848
Panel B: Manager-Loan-Level Portfolio Alignment				
Shared Loan/Issuer	0.001 (0.001)		0.001 (0.001)	
$-1 \times w - w^A $		-0.007 (0.024)		-0.003 (0.024)
Merger-Issuer/Loan FE	Yes	Yes	Yes	Yes
Manager Controls	Yes	Yes	Yes	Yes
Observations	624,782	624,782	823,190	823,190